JUN-GI JANG

jungijang@snu.ac.kr jungijang.github.io
Room 406, Building 43, 1 Gwanak-ro, Gwanak-gu, Seoul 08826, Republic of Korea

RESEARCH INTERESTS

Scalable Data Mining Systems, Tensor and Graph Mining, Model Efficiency and Compression, Applied AI

EXPERIENCE

Assistant Professor Sep. 2026 – Present
Seoul National University (SNU), Graduate School of Data Science
Joint Appointment, SNU College

Assistant Professor Sep. 2025 – Aug. 2026
Daegu Gyeongbuk Institute of Science & Technology (DGIST)
Department of Electrical Engineering and Computer Science

Postdoctoral Researcher Sep. 2023 – Aug. 2025
University of Illinois Urbana-Champaign (UIUC)
Advisor: Prof. Hanghang Tong

Postdoctoral Researcher Mar. 2023 – Aug. 2023
Seoul National University (SNU)
Advisor: Prof. U Kang

Research Intern Jul. 2020 – Aug. 2020
HYPERCONNECT

EDUCATION

Ph.D. in Computer Science and Engineering Mar. 2017 – Feb. 2023
Seoul National University
Thesis: Mining Real World Tensors via Efficient Tensor Decomposition Methods
Advisor: Prof. U Kang

B.S. in Mechanical and Aerospace Engineering Mar. 2010 – Feb. 2017
Seoul National University
Double major in Computer Science and Engineering

BEST PAPER AWARDS

Best Paper Award (Honorable Mention), ICDE May 2022
Best Paper Award (Best Research Paper), KDD Aug. 2021
Best Paper Award (1st Place), BigComp Jan. 2021

OTHER AWARDS AND HONORS

Outstanding Dissertation Award, SNU CSE Feb. 2023
SNU BK21 Star Researcher Award, SNU BK21 Feb. 2022
BK21 Best Graduate Student Award, SNU BK21 Feb. 2022
Future Gauss Lecture Award, Gauss Labs Feb. 2022
Naver Ph.D. Fellowship Award, Naver Dec. 2021

Qualcomm Innovation Fellowship, Qualcomm Nov. 2021
Yulchon AI Star Fellowship, Yulchon Foundation Sep. 2021
Humantech Paper Award (Honorable Mention, lead-author), Samsung Feb. 2018

REFEREED CONFERENCE AND JOURNAL PAPERS

28. **SCOUT: Coupling-Free Bounds for Trillion-Scale Top-k Retrieval in Sparse Tensor Factorization**
Jun-Gi Jang, Hyunsik Yoo, Ruizhong Qiu, Yinglong Xia, and Hanghang Tong
ACM SIGMOD/PODS International Conference on Management of Data (**SIGMOD**), 2027, Huntington Beach, CA, USA

27. **ProgNet: Program-Grounded Evidence Composition for Interpretable Graph Classification**
Minseok Jeon, Seunghyun Park, and **Jun-Gi Jang**
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2026, Jeju, South Korea

26. **DuET: Dual-View Tensor-to-Topology Spectral Adapter for Enhancing Sparse Tensor Factorization**
Jun-Gi Jang, Jingrui He, Andrew J. Margenot, and Hanghang Tong
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2026, Jeju, South Korea

25. **Fast and Accurate Domain Adaptation for Irregular and Regular Tensor Decomposition**
Junghun Kim, Ka Hyun Park, **Jun-Gi Jang**, and U Kang
IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 2026

24. **TUCKET: A Tensor Time Series Data Structure for Efficient and Accurate Factor Analysis over Time Ranges**
Ruizhong Qiu*, **Jun-Gi Jang***, Xiao Lin, Lihui Liu, and Hanghang Tong (* equal contribution)
International Conference on Very Large Data Bases (**VLDB**), 2025, London, United Kingdom

23. **Improving Group Fairness in Tensor Completion via Imbalance Mitigating Entity Augmentation**
Dawon Ahn*, **Jun-Gi Jang***, and Evangelos E. Papalexakis (* equal contribution)
The Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**), 2025, Sydney, Australia

22. **Fast and Accurate PARAFAC2 Decomposition for Time Range Queries on Irregular Tensors**
Jun-Gi Jang, Yong-chan Park, and U Kang
ACM International Conference on Information and Knowledge Management (**CIKM**), 2024, Boise, Idaho, USA
Oral presentation, acceptance rate 347/1496 ≈ 23%

21. **Compact Lossy Compression of Tensors via Neural Tensor-Train Decomposition**
Taehyung Kwon, Jihoon Ko, Jinhong Jung, **Jun-Gi Jang**, and Kijung Shin
Knowledge and Information Systems (**KAIS**), Oct. 2024

20. **Compact Decomposition of Irregular Tensors for Data Compression: From Sparse to Dense to High-Order Tensors**
Taehyung Kwon, Jihoon Ko, Jinhong Jung, **Jun-Gi Jang**, and Kijung Shin
ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2024, Barcelona, Spain
Oral presentation, acceptance rate ≈ 20%

19. **Fast and Accurate Domain Adaptation for Irregular Tensor Decomposition**
    Junghun Kim, Ka Hyun Park, **Jun-Gi Jang**, and U Kang
    ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2024, Barcelona, Spain
    Oral presentation, acceptance rate ≈ 20%
18. **Fast and Accurate Dual-Way Streaming PARAFAC2 for Irregular Tensors – Algorithm and Application**
    Jun-Gi Jang, Jeongyoung Lee, Yong-chan Park, and U Kang
    ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2023, Long Beach, CA, USA
    Oral presentation, acceptance rate 313/1416 ≈ 22.1%
17. **Accurate Open-set Recognition for Memory Workload**
    Jun-Gi Jang, Sooyeon Shim, Vladimir Egay, Jeeyong Lee, Jongmin Park, Suhyun Chae, and U Kang
    ACM Transactions on Knowledge Discovery from Data (**TKDD**), Jun. 2023
16. **Fast and Accurate Interpretation of Workload Classification Model**
    Sooyeon Shim, Doyeon Kim, **Jun-Gi Jang**, Suhyun Chae, Jeeyong Lee, and U Kang
    PLOS ONE, Mar. 2023
15. **Accurate Bundle Matching and Generation via Multitask Learning with Partially Shared Parameters**
    Hyunsik Jeon, **Jun-Gi Jang**, Taehun Kim, and U Kang
    PLOS ONE, Mar. 2023
14. **Static and Streaming Tucker Decomposition for Dense Tensors**
    Jun-Gi Jang and U Kang
    ACM Transactions on Knowledge Discovery from Data (**TKDD**), Feb. 2023
13. **Falcon: Lightweight and Accurate Convolution Based on Depthwise Separable Convolution**
    Jun-Gi Jang*, Chun Quan*, Hyun Dong Lee, and U Kang (* equal contribution)
    Knowledge and Information Systems (**KAIS**), Jan. 2023
12. **Accurate PARAFAC2 Decomposition for Temporal Irregular Tensors with Missing Values**
    Jun-Gi Jang, Jeongyoung Lee, Jiwon Park, and U Kang
    IEEE International Conference on Big Data (**BigData**), 2022, Osaka, Japan
    Oral presentation, acceptance rate 122/633 ≈ 19.2%
11. **DPar2: Fast and Scalable PARAFAC2 Decomposition for Irregular Dense Tensors**
    Jun-Gi Jang and U Kang
    IEEE International Conference on Data Engineering (**ICDE**), 2022, Virtual Event
    Oral presentation, acceptance rate 211/780 ≈ 27.1%
    🏆 **Best Paper Award, Honorable Mention**
10. **Large-Scale Tucker Tensor Factorization for Sparse and Accurate Decomposition**
    Jun-Gi Jang*, Moonjeong Park*, Jongwuk Lee, and Lee Sael (* equal contribution)
    The Journal of Supercomputing, May 2022
9. **Finding Key Structures in MMORPG Graph with Hierarchical Graph Summarization**
    Jun-Gi Jang, Chaeheum Park, Changwon Jang, Geonsoo Kim, and U Kang
    ACM Transactions on Knowledge Discovery from Data (**TKDD**), Feb. 2022
8. **Time-Aware Tensor Decomposition for Sparse Tensors**
    Dawon Ahn, **Jun-Gi Jang**, and U Kang
    Machine Learning, Sep. 2021

7. **Fast and Memory-Efficient Tucker Decomposition for Answering Diverse Time Range Queries**
   Jun-Gi Jang and U Kang
   ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2021, Virtual Event
   Oral presentation, acceptance rate 238/1541 ≈ 15.4%
   🏆 **Best Paper Award, Best Research Paper**
6. **Fast and Accurate Partial Fourier Transform for Time Series Data**
   Yong-chan Park, **Jun-Gi Jang**, and U Kang
   ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD**), 2021, Virtual Event
   Oral presentation, acceptance rate 238/1541 ≈ 15.4%
5. **VEST: Very Sparse Tucker Factorization of Large-Scale Tensors**
   Moonjeong Park*, **Jun-Gi Jang***, and Lee Sael (* equal contribution)
   IEEE International Conference on Big Data and Smart Computing (**BigComp**), 2021, Online
   🏆 **Best Paper Award, 1st Place**
4. **D-Tucker: Fast and Memory-Efficient Tucker Decomposition for Dense Tensors**
   Jun-Gi Jang and U Kang
   IEEE International Conference on Data Engineering (**ICDE**), 2020, Online
   Short paper, acceptance rate ≈ 32%
3. **S3CMTF: Fast, Accurate, and Scalable Method for Incomplete Coupled Matrix-Tensor Factorization**
   Dongjin Choi, **Jun-Gi Jang**, and U Kang
   PLOS ONE, Jun. 2019
2. **High-Performance Tucker Factorization on Heterogeneous Platforms**
   Sejoon Oh, Namyong Park, **Jun-Gi Jang**, Lee Sael, and U Kang
   IEEE Transactions on Parallel and Distributed Systems (**TPDS**), Apr. 2019
1. **Zoom-SVD: Fast and Memory Efficient Method for Extracting Key Patterns in an Arbitrary Time Range**
   Jun-Gi Jang, Dongjin Choi, Jinhong Jung, and U Kang
   ACM International Conference on Information and Knowledge Management (**CIKM**), 2018, Turin, Italy
   Oral presentation, acceptance rate 147/826 ≈ 17.8%

TEACHING EXPERIENCE

Instructor, Big Data and Knowledge Management Systems 2, SNU	Fall 2026
Instructor, BE202 Introduction to Data Science, DGIST	Spring 2026
Co-Instructor, TM545c Fundamentals of Data Science, DGIST	Spring 2026
Lead T.A., M2177.004900 Theory and Lab of IoT, AI, and Big Data, SNU	Spring 2020
T.A., M2177.004900 Theory and Lab of IoT, AI, and Big Data, SNU	Fall 2019
T.A., M2177.004900 Theory and Lab of IoT, AI, and Big Data, SNU	Spring 2019
T.A., M1522.001400 Introduction to Data Mining, SNU	Spring 2018
T.A., M1522.000900 Data Structure, SNU	Fall 2017

INVITED TALKS

Data Intelligence Workshop, DASFAA — Apr. 2026
The Future of Data Workshop 2023, KCC DB Society, KIISE — Jun. 2023
Korea Computer Congress 2022, KIISE — Jun. 2022
Korea Software Congress 2021, KIISE — Dec. 2021
Regular Seminar, Qatar Computing Research Institute (QCRI) — Sep. 2021
Korea Computer Congress 2020, KIISE — Jul. 2020
Korea Software Congress 2018, KIISE — Dec. 2018
Samsung AI Forum, Samsung — Sep. 2018

PROFESSIONAL SERVICES

Workshop Organizing Committee

Workshop on Interplay Between Classical Tensor Methods and Foundation Models, KDD 2026

Area Chair

KDD 2026 – 2027

Session Chair

DASFAA 2026

PC Member and Reviewer

KDD	2023 – 2026	TKDE journal	2024
CIKM	2025 – 2026	TSP journal	2024
WWW	2026 – 2027	Machine Learning journal	2023 – 2024
AAAI	2024 – 2027	TIST journal	2023
BigData	2026	TPDS journal	2023
SDM	2024	DAMI journal	2023, 2026
KAIS journal	2024		

External Reviewer

TKDD	2025	ICLR	2021
NAACL	2024	NeurIPS	2020 – 2022
DSAA	2024	CIKM	2018 – 2019
KDD	2019 – 2022	ICDM	2018
WWW	2019 – 2021	WSDM	2018